

Recall from last class:

- A linear system is called **homogeneous** if it can be written as $A\vec{x} = \vec{0}$ where A is an $m \times n$ matrix, $\vec{x} \in \mathbb{R}^n$, and $\vec{0}$ is the vector of m zeros.
- **Theorem:** If $A\vec{x} = \vec{b}$ is consistent for a given $\vec{b}$, and $\vec{p}$ is a solution, then the solution set of $A\vec{x} = \vec{b}$ is the set of vectors of the form $\vec{p} + \vec{v}_h$ where $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$.

Definition 1. A set of vectors $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

1. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent?

Let $A = [\vec{a}_1 \dots \vec{a}_n]$, then $\{\vec{a}_1 \dots \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

2. Is a set with only one vector always linearly independent?

3. If $\vec{0} \in \{\vec{v}_1, \dots, \vec{v}_p\}$, then is the set linearly dependent?

4. If $\vec{0} \in \text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$, then the vectors are linearly independent?

5. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. Are the vectors linearly independent?

Theorem 7: A set $S = \{\vec{v}_1, \dots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_{j-1}$.

6. Prove Theorem 7.

7. Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$

(a) Describe $\text{Span}\{\vec{u}, \vec{v}\}$

(b) Prove $\vec{w} \in \text{Span}\{\vec{u}, \vec{v}\}$ if and only if $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent.

Theorem 8: Any set $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

8. Prove the theorem.